

Redrob Recruiter Matcher AI

Senior AI Engineer Search — Intelligent Ranking & Behavioral Re-Ranking

Team Name: Resume Run

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OS Platform: Linux (Ubuntu 24.04 LTS)

Execution Mode: 100% Offline, Zero APIs, Zero Network Calls

The Recruiting Dilemma: Beyond Keyword Matching

Traditional ATS keyword-matching filters fail to find the right candidates. Recruiters are either swamped with irrelevant keyword-stuffed resumes or miss top-tier candidates who don't list specific brand names. Our solution ranks candidates the way a human recruiter would: *assessing context, experience depth, and real-world availability.*

Core Design Principles:

- **Rigorous Technical Gating:** Calculate actual years of AI/ML-specific experience rather than relying on profile-wide metrics.
- **Disqualification of Profile Honeypots:** Identify and immediately filter logical inconsistencies (e.g. 0-duration expert skills or impossible startup timelines).
- **Filtering Out Non-Product Gaps:** Reject candidates with experiences exclusively in mass consulting and IT-service firms.
- **Behavioral Modulation:** Use real-time availability metrics (notice period, response speed, activity, location) as a tie-breaker, not a replacement for talent.

Unmasking profile traps and honeypots

Analyzing 100,000 candidates revealed intentional traps designed to trigger simple keyword matchers:

Traps Identified	System Defense	Impact
Rule 1: Expert Skill, 0 Duration Candidates claiming 'expert' or 'advanced' proficiency in AI tools but listing 0 months of experience.	Identified in 21 profiles. Immediate disqualification (Score set to 0.0).	Blocks keyword-stuffers.
Rule 2: Impossible Foundings Candidates claiming employment at startups (Krutrim, Sarvam AI) before those companies were founded (pre-2023).	Identified in 73 profiles. Immediate disqualification (Score set to 0.0).	Blocks fraudulent histories.
Rule 3: Consulting Only Candidates whose entire work history is at services firms (TCS, Infosys, Wipro, etc.).	Checks entire career history. Immediate disqualification (Score set to 0.0).	Ensures product-native engineering talent.

Scoring & Re-Ranking Architecture

Our ranking algorithm computes a candidate's final score by multiplying the **Technical Profile Fit** with the **Behavioral Availability Modifier**.

Technical Fit Component (Weight)	Description / Formula	Behavioral Modifier (Range)	Description / Logic
ML Experience Duration (25%)	Months in roles matching ML keywords or description terms. Prefers 3+ years.	Recency & Activity	Bonus for active, penalty for inactivity > 6 months (0.4x).
Current Title Fit (20%)	1.0 for ML/AI engineers; 0.9 for Data Scientists; 0.8 for Backend w/ ML; 0.0 for non-tech.	Notice Period	Bonus for <=30 days, penalty for >90 days (0.5x).
Core Skills Alignment (20%)	Core skills (Embeddings, RAG, LoRA, Search) scaled by proficiency & log-scaled duration.	Location/Relocation	Bonus for NCR/Pune; penalty if non-local and not willing to relocate (0.5x).
Experience Years (15%)	Flat 1.0 for 5-9 years; drops linearly for junior and senior outliers.	Recruiter Response	Bonus for fast, responsive candidates; penalty for response rate < 20% (0.7x).
Product Fit / Keywords (20%)	Ratio of months at product firms (10%) + occurrence of search/vector terms in job history (10%).	GitHub Activity	Small bonus (+3%) for active GitHub contributors.

Validation Results & High-Performance Specs

Output Quality and Schema Validation:

- **Schema Compliant:** Generated CSV file *team_run.csv* conforms to the exact headers (candidate_id, rank, score, reasoning) required and passed the validator check.
- **No Formatting Artifacts:** All generated reasoning statements are clean, use natural phrasing, do not contain trailing double periods, and display proper spacing.
- **100% Genuine ML Shortlist:** The top-ranked candidates consist entirely of high-caliber NLP, AI, and Machine Learning engineers with years of product experience, successfully filtering out QA and frontend profiles.

Compute and Speed Performance:

- **Fast Execution:** Scores and ranks 100,000 profiles in **under 13 seconds** (averaging 12.26s) on a single CPU core, utilizing vector/matrix logic in NumPy.
- **Minimal Memory Footprint:** Fits entirely within standard RAM using streaming file reading (JSONL), leaving 99% of resources free.
- **Air-Gapped Compliance:** Runs entirely offline, requiring zero network calls or external APIs, securing candidate privacy.

Conclusion: Our rule-based system delivers a highly-focused, premium shortlist of vetted candidates. By coupling strict technical gating with behavioral modulation, it mirrors the selection quality of an elite recruiter.